Computer practical multi-state models

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Abstract

The objective of this computer practical is to get to know **mstate** through a worked out research question in a multi-state model. No familiarity with **mstate** is assumed; the R code to perform the analysis is given. Questions are posed to reflect on what you are doing and to study output and results in more detail.

Keywords: .

concordance:Mstate$_p$*ractical.tex* : *Mstate*$_p$*ractical.Rnw* : 1311413211210113012211212011120121112701231
3172011210311240124112701160118012911210111101116022101130121 11−31740112104190122112110122211
3171711210111210111401224112401231128101241121011190123112401 2111−317181124012111−
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1. Introduction

The data to be studied in this computer practical is from the European Blood and Marrow Transplantation (EBMT) registry. It contains data on 1977 patients who received an allogeneic bone marrow transplantation. Time is measured in days from transplantation. The only prognostic factor is the EBMT risk score, called `score` in this data, which divides patients into three risk groups, low risk, medium risk and high risk patients. This score is a combination of donor-recipient match and age of the recipient, among other things. It is generally accepted as being highly predictive of both relapse and survival before and after relapse. The study question is about the impact of a relapse on survival. It is well known that over the last years (decades) the treatment of relapse has been improved, among other things. As a result, a relapse is not considered as serious as it used to be say 10, 15 years ago. Some clinicians have gone so far as to suggest that having a relapse does not influence mortality anymore. The idea of this practical is to study these questions.

2. A first look at the data

The data are part of the **mstate** package. Make sure that you have installed **mstate**. We load the package and read in the data.

```
> library(mstate)
```

```
> data(ebmt1)
```

Take a look at the data. The functions *head* and *tail*, showing the first and last six lines of the dataset, are quite useful for larger datasets.

```
> head(ebmt1)

  patid  srv srvstat  rel relstat      yrel age       score
1     1 1610       0 1610       0      <NA>  28 Medium risk
2     2  961       1  165       1 1997-1999  33 Medium risk
3     3 1508       0 1508       0      <NA>  38 Medium risk
4     4 1376       0 1376       0      <NA>  15 Medium risk
5     5 1585       0  133       1 1997-1999  26 Medium risk
6     6  190       1   63       1 1997-1999  22 Medium risk
```

```
> tail(ebmt1)

     patid srv srvstat rel relstat      yrel age       score
1972  1972 746       1 746       0      <NA>  47 Medium risk
1973  1973 123       1 122       1 1997-1999  44   High risk
1974  1974 632       1 632       0      <NA>  32    Low risk
1975  1975 911       0 215       1     2000-  39 Medium risk
1976  1976  78       1  78       0      <NA>  42 Medium risk
1977  1977 180       1 180       0      <NA>  19 Medium risk
```

The function *dim* can be used to find the number of rows (patients in this case) and columns (variables) in the data.

```
> dim(ebmt1)

[1] 1977    8
```

Take a look at the structure of the data. Type `help(ebmt1)` to get information on the variables and their meaning and values. We will now have a closer look at the events in the data. To see how many events have occurred, the function *table* can be used:

```
> table(ebmt1$srvstat)

   0    1
1105  872
```

```
> table(ebmt1$relstat)

   0    1
1521  456
```

```
> table(ebmt1$srvstat, ebmt1$relstat)
```

```
    0   1
0 836 269
1 685 187
```

3. Standard survival analysis and competing risks

3.1. Overall survival

Before formulating a multi-state model to study the data let us simplify matters first and look at overall survival. The R library **survival** has got special functions to obtain Kaplan-Meier survival curves, most notably the function *survfit*.

```
> library(survival)
> sf0 <- survfit(Surv(srv, srvstat) ~ 1, data=ebmt1)
```

The result is a so called `survfit` object, for which a number of functions (methods is the better word) are defined. One of those is the *print* method. If you just type in the name of the object at the command line, the *print* method is invoked.

```
> sf0

Call: survfit(formula = Surv(srv, srvstat) ~ 1, data = ebmt1)

      n  events  median 0.95LCL 0.95UCL
   1977     872    2537    1828      NA
```

The *summary* method is useful to obtain Kaplan-Meier estimates of the survival probabilities (and the numbers at risk) at selected time points of interest. Let's have a look at 0 ... 7 years.

```
> year <- 365.25
> summary(sf0, times=year*(0:7))

Call: survfit(formula = Surv(srv, srvstat) ~ 1, data = ebmt1)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
    0   1977       0    1.000  0.0000        1.000        1.000
  365   1199     698    0.640  0.0109        0.619        0.662
  730    951      88    0.591  0.0113        0.570        0.614
 1096    679      48    0.558  0.0116        0.535        0.581
 1461    454      15    0.544  0.0119        0.521        0.568
 1826    273      14    0.524  0.0126        0.500        0.550
 2192    148       5    0.513  0.0133        0.488        0.540
 2557     59       3    0.497  0.0162        0.466        0.529
```

And the results can be plotted as well; I have used *xscale* here to plot the results in years, rather than days. Figure 1 shows the result.

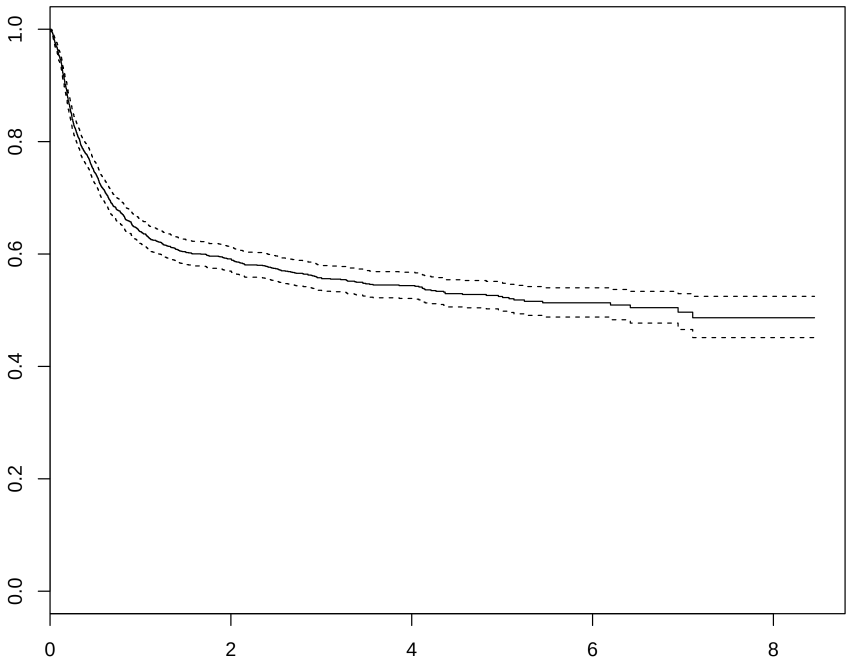

Figure 1: The Kaplan-Meier curve for overall survival

```
> plot(sf0, xscale=year)
```

3.2. Competing risks

The next step is to look at the two competing risks outcomes, namely relapse and death before relapse, also called non-relapse mortality (NRM). The composite endpoint that consists of either relapse or non-relapse mortality is called relapse-free survival, RFS in short. The data does not yet have these time and status variables, nor the ones for competing risks, so we have to compute them first. We will denote the time (days since transplantation) of occurrence of the first of relapse or death `rfs`, the status variable for RFS `rfsstat`, and the competing risks variable `rfscr`, with values 0 for censored, 1 for relapse, and 2 for NRM.

Question 1. Think how you would go about computing these variables, before looking at the code below.

```
> ebmt1$rfs <- pmin(ebmt1$rel, ebmt1$srv)
> ebmt1$rfsstat <- 1 - (1-ebmt1$relstat) * (1-ebmt1$srvstat)
> ebmt1$rfscr <- ebmt1$relstat
> ebmt1$rfscr[ebmt1$relstat==0 & ebmt1$srvstat==1] <- 2
> ebmt1$rfscr <- factor(ebmt1$rfscr, levels=0:2,
+                       labels=c("Censored", "Relapse", "NRM"))
```

To check whether we have coded correctly, we can make frequency tables of `rfsstat` and `rfscr`, and a cross-table of `rel` and `srv`.

```
> table(ebmt1$rfsstat)

   0    1
 836 1141

> table(ebmt1$rfscr)

Censored  Relapse      NRM
     836      456      685

> table(ebmt1$srvstat, ebmt1$relstat)

      0   1
  0 836 269
  1 685 187
```

Question 2. Check whether these numbers make sense.

If you call *survfit* with a status variable that is a factor, the software will assume that it concerns a competing risks outcome, where the first level is taken to be the censoring. Here is how it works (also the summary function works similarly).

```
> sfcr <- survfit(Surv(rfs, rfscr) ~ 1, data=ebmt1)
> sfcr

Call: survfit(formula = Surv(rfs, rfscr) ~ 1, data = ebmt1)

           n nevent    rmean*
Relapse 1977    456  707.3375
NRM     1977    685 1046.9510
        1977      0 1333.7115
   *mean time in state, restricted (max time = 3088 )

> summary(sfcr, times=year*(0:7))

Call: survfit(formula = Surv(rfs, rfscr) ~ 1, data = ebmt1)

 time n.risk n.event P(Relapse) P(NRM)   P()
    0   1977       0      0.000  0.000 1.000
  365   1012     897      0.156  0.306 0.538
  730    762     127      0.197  0.335 0.468
 1096    518      68      0.226  0.352 0.422
 1461    333      22      0.245  0.354 0.401
 1826    198      16      0.258  0.363 0.378
 2192    101       7      0.271  0.367 0.361
 2557     40       3      0.280  0.371 0.348

> plot(sfcr, lwd=2, col=1:2, xscale=year)
> legend("bottomright", c("Relapse", "NRM"), lwd=2, col=1:2, bty="n")
```

Question 3. We have already calculated time and status indicator for RFS, namely `rfs` and `rfsstat`. Using the same methods as for overall survival, calculate and make a plot of the Kaplan-Meier curve for RFS. Ask for the probabilities at each year, until 7 years. Check your results with the competing risks results.

Question 4. Check your results also with the Kaplan-Meier curves for overall survival. Which one is higher? What does the difference represent?

4. Multi-state model

4.1. Defining the multi-state model

All this doesn't take into account the effect of relapse on survival. We will do that now using a multi-state model with three states, Transplant (T), state 1, Relapse (R), state 2, and Death (D), state 3, and with 3 transitions, the first being T $\rightarrow$ R, the second T $\rightarrow$ D, the third R $\rightarrow$ D using the **mstate** R package.

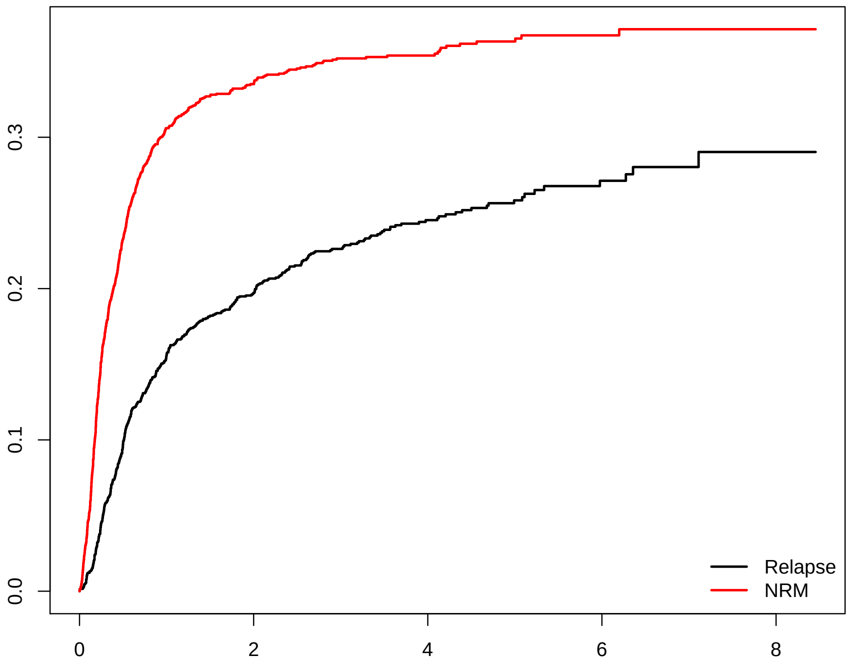

Figure 2: Cumulative incidence curves for relapse and non-relapse mortality

Question 5. Make a drawing for yourself of the multi-state model, indicating the state numbers and names, and the transitions indicating the transition numbers. Using the results of Sections 2 and 3, figure out ow many events have occurred for each transition, and write them in your diagram.

In **mstate**, such a multi-state model is specified with a 3×3 transition matrix (in general, $S \times S$, with S the number of states in the multi-state model). Consecutive numbers 1 through K designate the allowed direct transitions in the multi-state model (in our case $K = 3$). If the (i, j) entry of this matrix equals k, then the kth transition runs from state i to state j. If a direct transition is not allowed, an `NA` is used in the transition matrix. In our case the multi-state model is specified as the transition matrix

```
> tmat <- matrix(NA,3,3)
> tmat[1,2:3] <- 1:2
> tmat[2,3] <- 3
> dimnames(tmat) <- list(c("T","R","D"), c("T","R","D"))
> tmat
```

```
   T  R  D
T NA  1  2
R NA NA  3
D NA NA NA
```

Because illness-death models are a common type of multi-state models, a function has been defined within **mstate** to quickly construct a transition matrix for it.

```
> trans.illdeath(names=c("T","R","D"))
```

```
    to
from  T  R  D
   T NA  1  2
   R NA NA  3
   D NA NA NA
```

4.2. Data preparation

Now to create a long format dataset for flexible multi-state modeling, we use the function *msprep*. The function *msprep* has *time* and *status* arguments which can be specified as $n \times S$ matrices or data frames containing the arrival times and status indicators for each of the S states of each of the n individuals. Alternatively, character vectors of length S can be specified, containing the columns names of these times and status indicators, plus a *data* argument, specifying the data frame containing these columns. Also a *trans* argument specifying the form of the multi-state model through the transition matrix, needs to be specified. Optionally, a vector of covariate names may be specified through the argument `keep`. Below we will specify *time* and *status* through character vectors. The `time` and `status` variables for relapse and survival are already available in the data, but the time of transplant (state 1) is not. Since each individual starts in this state, it is irrelevant what is specified there; in such a case `NA` may be specified as column names in the `time` and `status` arguments.

Now we are set to go. We will call *msprep*:

```
> covs <- c("score","yrel")
> msebmt <- msprep(
+      time=c(NA,"rel","srv"),
+      status=c(NA,"relstat","srvstat"),
+      data=ebmt1, trans=tmat, id="patid",
+      keep=covs)
```

The order of the elements of the `time` and `status` arguments has to be the same as the order of the states specified in the transition matrix. Another optional argument to *msprep* is `start`, a list with elements state and time, containing starting states and times of the subjects in the data. The default of `start` is NULL, in which case it is assumed that all subjects start in state 1 at time 0. The result of *msprep* is an object of class `"msdata"`, which is simply a data frame with the transition matrix as an attribute.

```
> head(msebmt)

An object of class 'msdata'

Data:
  patid from to trans Tstart Tstop time status       score      yrel
1     1    1  2     1      0  1610 1610      0 Medium risk      <NA>
2     1    1  3     2      0  1610 1610      0 Medium risk      <NA>
3     2    1  2     1      0   165  165      1 Medium risk 1997-1999
4     2    1  3     2      0   165  165      0 Medium risk 1997-1999
5     2    2  3     3    165   961  796      1 Medium risk 1997-1999
6     3    1  2     1      0  1508 1508      0 Medium risk      <NA>
```

Question 6. Compare the data of the first two patients with those in the original, wide format, data, and look whether this makes sense to you.

The number of transitions is counted using the function *events*.

```
> events(msebmt)

$Frequencies
    to
from    T    R    D no event total entering
   T    0  456  685      836           1977
   R    0    0  187      269            456
   D    0    0    0      872            872

$Proportions
    to
```

```
from         T         R         D  no event
   T 0.0000000 0.2306525 0.3464846 0.4228629
   R 0.0000000 0.0000000 0.4100877 0.5899123
   D 0.0000000 0.0000000 0.0000000 1.0000000
```

Question 7. Check whether these are the same numbers as obtained in Section 2.

5. Non-parametric estimation

5.1. Cumulative hazards

Even though there are no covariates involved, we will use Cox models, "empty" Cox models in the sense that no covariates are included in the formula, only the `strata`. The first model that we will fit is a Cox model, stratified according to `trans`.

```
> c0 <- coxph(Surv(Tstart,Tstop,status) ~ strata(trans),
+     data=msebmt)
> c0
```

```
Call:  coxph(formula = Surv(Tstart, Tstop, status) ~ strata(trans),
    data = msebmt)

Null model
  log likelihood= -9036.042
  n= 4410
```

The cumulative hazards for the three transitions can be plotted using *survfit* (results not shown).

```
> plot(survfit(c0), fun="cumhaz", col=1:3, xscale=year)
```

Alternatively, the same can be obtained using *msfit* from **mstate**. The advantage is an automatic legend with the transitions, and especially that the result can be used later for estimation of state occupation and transition probabilities. The call to estimate the transition hazards is given below.

```
> msf0 <- msfit(c0, trans=tmat)
```

We will look at the structure of the resulting `msfit` object later, in Section 6. It has a *plot* method which plots the cumulative transition hazards. The result is shown in Figure 3.

```
> plot(msf0)
```

Question 8. Study and comment on the shape and height of the three cumulative hazards.

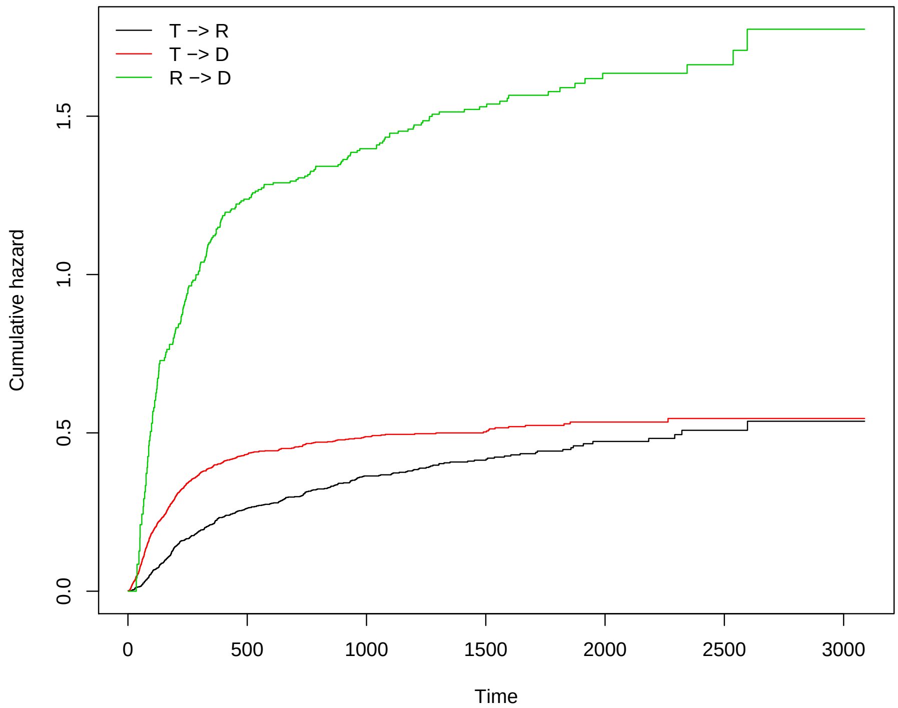

Figure 3: Non-parametric estimates of the cumulative hazards

We will look now at a model where the baseline transition intensities into death are assumed to be proportional. Thus, the transitions 2 and 3 share a common baseline. We do this by defining a time-dependent covariate `rel.surv` which is 0 for transition 2 and 1 for transition 3, so it is 0 if no relapse has occurred and 1 if a relapse has occurred. It measures the effect of relapse on survival. The whole model can be fitted using a stratified Cox regression model with strata defined by the receiving state (`to` in the data), and with `rel.srv` as additional covariate.

```
> msebmt$rel.srv <- 0
> msebmt$rel.srv[msebmt$trans==3] <- 1
> c1 <- coxph(Surv(Tstart,Tstop,status) ~ rel.srv +
+     strata(to), data=msebmt)
> c1

Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ rel.srv + strata(to),
    data = msebmt)

          coef exp(coef) se(coef)    z      p
rel.srv 1.1355    3.1126   0.0892 12.7 <2e-16

Likelihood ratio test=136.3  on 1 df, p=<2e-16
n= 4410, number of events= 1328
```

Question 9. What does the hazard ratio of `rel.srv` imply?

Question 10. What is the difference between this model and a time-dependent Cox model?

In Section 6 we will see how the cumulative hazards of this model can be obtained; for now we will continue with estimation of the transition probabilities for the first model.

5.2. Transition probabilities

The result of *`msfit`* can be used as input for *`probtrans`*. Further arguments to *`probtrans`* are `predt`, the time of prediction, and `direction`, specifying the direction of the prediction, either `"forward"` or `"fixedhorizon"`. The output of *`probtrans`* is a list with S elements (S being the number of states) with element `[[g]]` containing estimates of either $P_{gh}(\texttt{predt}, t)$ for all event times $t >$ `predt` in case of forward prediction, or $P_{gh}(s, \texttt{predt})$ for all event times $s <$ `predt` in case of fixed horizon prediction. The standard errors of these prediction probabilities are also given. We will call *`probtrans`* with `msf0` as input.

```
> pt0 <- probtrans(msf0, predt=0)
```

The function *`probtrans`* returns an object of class `"probtrans"`. For such objects a *`summary`* method has been defined, which shows heads and tails of predictions from each starting state.

```
> summary(pt0)
```

```
An object of class 'probtrans'

Prediction from state 1 (head and tail):

  time   pstate1     pstate2     pstate3         se1          se2
1  0.0 1.0000000 0.000000000 0.000000000 0.000000000 0.0000000000
2  0.5 0.9979762 0.001011890 0.001011890 0.001009842 0.0007140661
3  1.0 0.9974704 0.001517706 0.001011890 0.001128866 0.0008746228
4  2.0 0.9969646 0.001517706 0.001517706 0.001236381 0.0008746228
5  5.0 0.9954464 0.001517706 0.003035927 0.001513267 0.0008746228
6  7.0 0.9944345 0.001517706 0.004047816 0.001672198 0.0008746228
           se3
1 0.0000000000
2 0.0007140661
3 0.0007140661
4 0.0008746227
5 0.0012363792
6 0.0014270430

...

    time   pstate1   pstate2   pstate3        se1        se2
598 2322 0.3481461 0.1601459 0.4917080 0.01498317 0.01308607
599 2344 0.3481461 0.1558176 0.4960363 0.01498317 0.01341077
600 2537 0.3481461 0.1487350 0.5031189 0.01498317 0.01447679
601 2596 0.3481461 0.1388193 0.5130346 0.01498317 0.01637722
602 2597 0.3381990 0.1487664 0.5130346 0.01747057 0.01888280
603 3088 0.3381990 0.1487664 0.5130346 0.01747057 0.01888280
           se3
598 0.01362048
599 0.01413516
600 0.01545408
601 0.01762046
602 0.01762046
603 0.01762046

Prediction from state 2 (head and tail):

  time pstate1 pstate2 pstate3 se1 se2 se3
1  0.0       0       1       0   0   0   0
2  0.5       0       1       0   0   0   0
3  1.0       0       1       0   0   0   0
4  2.0       0       1       0   0   0   0
5  5.0       0       1       0   0   0   0
6  7.0       0       1       0   0   0   0

...
```

```
    time pstate1   pstate2   pstate3 se1        se2        se3
598 2322       0 0.1922028 0.8077972   0 0.02959181 0.02959181
599 2344       0 0.1870082 0.8129918   0 0.02923229 0.02923229
600 2537       0 0.1785078 0.8214922   0 0.02905933 0.02905933
601 2596       0 0.1666073 0.8333927   0 0.02930826 0.02930826
602 2597       0 0.1666073 0.8333927   0 0.02930826 0.02930826
603 3088       0 0.1666073 0.8333927   0 0.02930826 0.02930826

Prediction from state 3 (head and tail):

  time pstate1 pstate2 pstate3 se1 se2 se3
1  0.0       0       0       1   0   0   0
2  0.5       0       0       1   0   0   0
3  1.0       0       0       1   0   0   0
4  2.0       0       0       1   0   0   0
5  5.0       0       0       1   0   0   0
6  7.0       0       0       1   0   0   0

...

    time pstate1 pstate2 pstate3 se1 se2 se3
598 2322       0       0       1   0   0   0
599 2344       0       0       1   0   0   0
600 2537       0       0       1   0   0   0
601 2596       0       0       1   0   0   0
602 2597       0       0       1   0   0   0
603 3088       0       0       1   0   0   0
```

Clearly, the probability of being in state 1 is zero now for predictions from state 2. For the predictions from state 3, we see that only `pstate3`, the probability of being in state 3, is 1. These prediction probabilities can also be accessed more directly. For instance, predictions from state 1 are accessed with

```
> pt01 <- pt0[[1]]
> head(pt01)

  time   pstate1     pstate2     pstate3         se1          se2
1  0.0 1.0000000 0.000000000 0.000000000 0.000000000 0.0000000000
2  0.5 0.9979762 0.001011890 0.001011890 0.001009842 0.0007140661
3  1.0 0.9974704 0.001517706 0.001011890 0.001128866 0.0008746228
4  2.0 0.9969646 0.001517706 0.001517706 0.001236381 0.0008746228
5  5.0 0.9954464 0.001517706 0.003035927 0.001513267 0.0008746228
6  7.0 0.9944345 0.001517706 0.004047816 0.001672198 0.0008746228
           se3
1 0.0000000000
2 0.0007140661
```

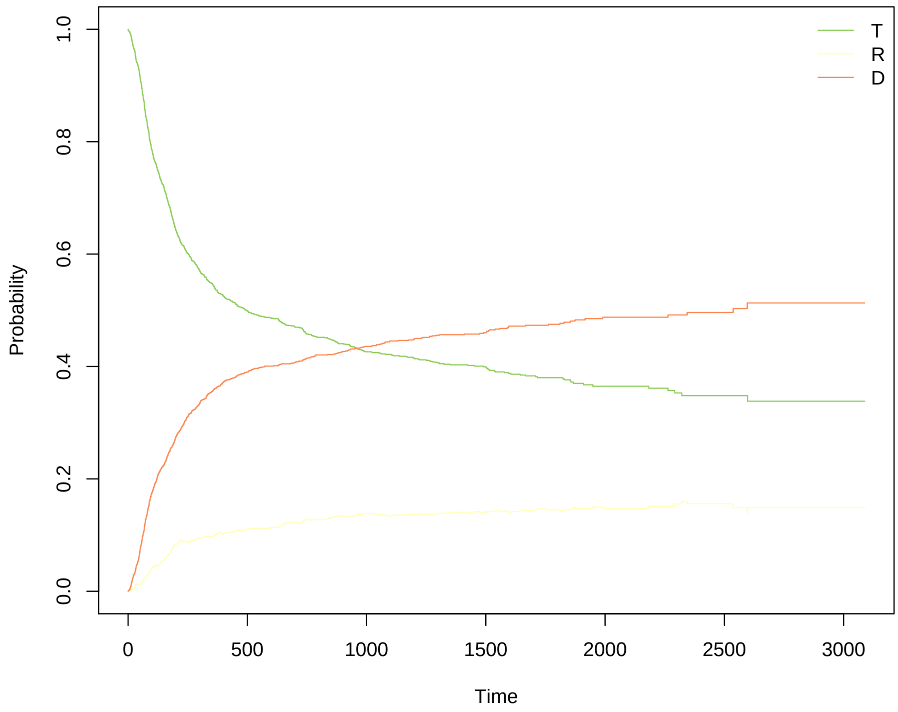

Figure 4: Ordinary plot of state probabilities

```
3 0.0007140661
4 0.0008746227
5 0.0012363792
6 0.0014270430
```

Also a *plot* method has been defined for "probtrans" objects. The argument type distinguishes between different types of plots (have a look at the help function for more information). The simplest is using "single", as shown in Figure 4.

```
> plot(pt0, type="single")
```

The probabilities shown in Figure 4 are the probabilities of being in state 1, 2, and 3, given that the patient was in state 1 (that is the default of the *plot* method of *probtrans*) at time 0 (that was specified as predt in the call to *probtrans*). Since in this case all patients started in state 1, these probabilities coincide with the *state occupation probabilities*.

Perhaps more interesting plots are obtained using stacked probability plots. In such plots, the distance between any two curves represents the probability of being in the corresponding state. For our three states, we get a nice picture if we stack first the probability of a relapse, then death, finally transplant. The order relapse, death, transplant is not the same as the states defined in tmat, therefore we specify the argument ord as c(2,3,1) to indicate that

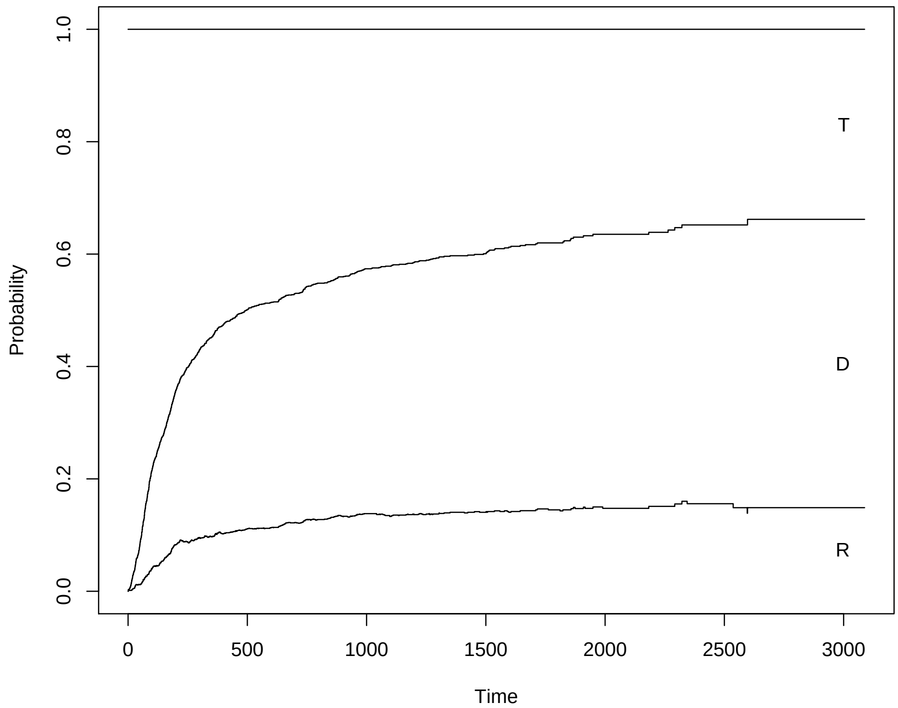

Figure 5: Stacked prediction probabilities from state 1 at time=0

we want to stack the probabilities of first state 2 (relapse), then state 3 (death), then state 1 (transplant). The result is shown in Figure 5.

```
> plot(pt0, ord=c(2,3,1))
```

The lowest curve of Figure 5 indicates the probability of being in state 2 (alive with relapse), the second curve the probability of being in state 2 plus the probability of being in state 3. The distance between these two curves is the probability of dying. The second curve thus indicates the probability of treatment failure (either relapse or death). The distance between the second curve and 1 is relapse-free survival.

Question 11. The lowest curve in Figure 5 increases most of the time (though at a slower rate after one year). Occasionally, if you look closely, you will see a small decrease. How is that possible?

An even nicer picture is obtained when specifying "filled" for the type argument. In that case the spaces between two adjacent curves are colored. Figure 6 shows an example.

```
> plot(pt0, ord=c(2,3,1), type="filled")
```

Question 12. Figure out and write down the state occupation probabilities at 5 years.

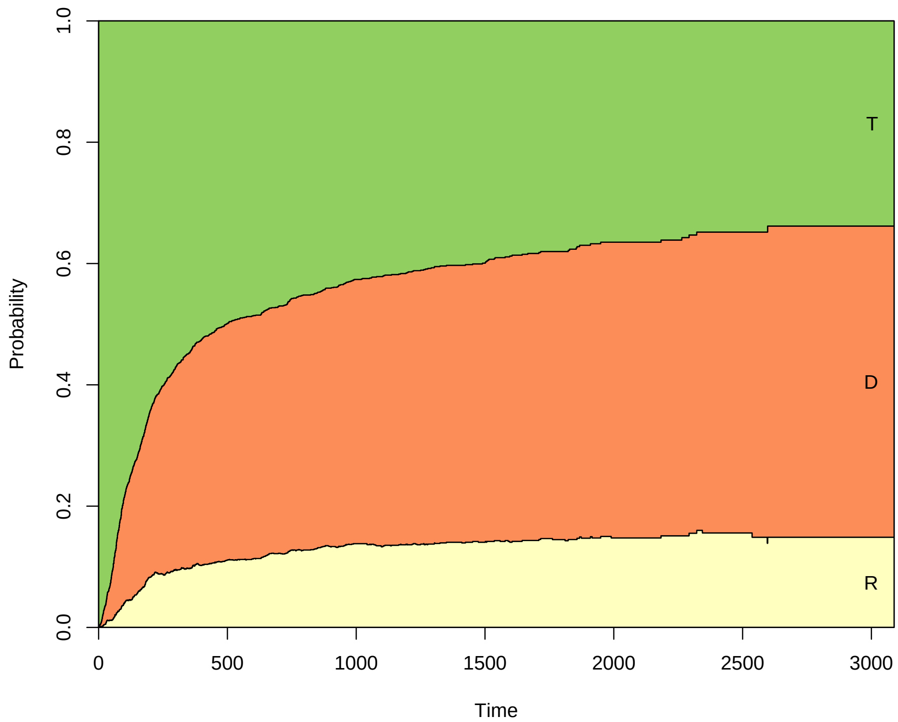

Figure 6: Stacked (and filled) prediction probabilities from state 1 at time=0

Figure 6 shows the probability of being in state 3 (death) over time. Of the patients that died, some died directly, others after relapse. The current analysis does not distinguish between these possibilities. This can however be incorporated in a quite simple way, by defining a different multi-state model, and from that proceeding in the same way as before.

```
> tmat4 <- transMat(x = list(c(2, 3), c(4), c(), c()),
+                   names = c("Tx", "Rel", "NRM", "Death after Rel"))
> tmat4

                 to
from              Tx Rel NRM Death after Rel
  Tx              NA   1   2              NA
  Rel             NA  NA  NA               3
  NRM             NA  NA  NA              NA
  Death after Rel NA  NA  NA              NA

> msebmt4 <- msprep(
+     time = c(NA, "rel", "srv"),
+     status = c(NA, "relstat", "srvstat"),
+     data=ebmt1, trans=tmat, id="patid",
+     keep=covs)
> events(msebmt4)

$Frequencies
    to
from    T    R    D no event total entering
   T    0  456  685      836           1977
   R    0    0  187      269            456
   D    0    0    0      872            872

$Proportions
    to
from         T         R         D  no event
   T 0.0000000 0.2306525 0.3464846 0.4228629
   R 0.0000000 0.0000000 0.4100877 0.5899123
   D 0.0000000 0.0000000 0.0000000 1.0000000

> c0 <- coxph(Surv(Tstart,Tstop,status) ~ strata(trans),
+     data=msebmt4)
> c0

Call:  coxph(formula = Surv(Tstart, Tstop, status) ~ strata(trans),
    data = msebmt4)

Null model
  log likelihood= -9036.042
  n= 4410
```

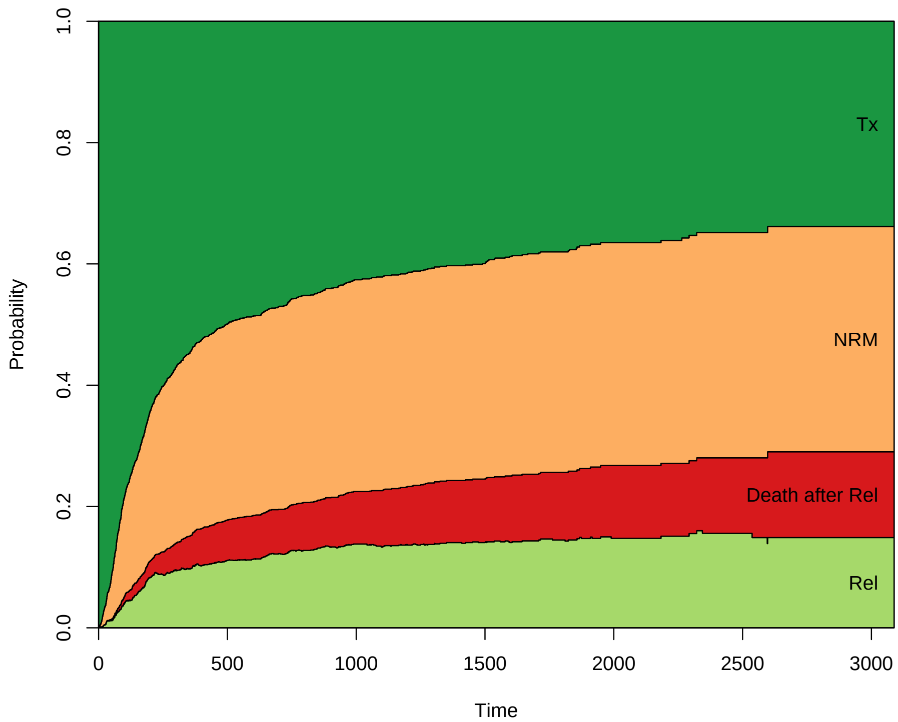

Figure 7: Stacked (and filled) prediction probabilities from state 1 at time=0

```
> msf0 <- msfit(c0, trans=tmat4)
> # plot(msf0) # Try this yourself
> pt0 <- probtrans(msf0, predt=0)

> plot(pt0, ord=c(2, 4, 3, 1), type="filled")
```

Question 13. Figure out and write down the state occupation probabilities at 5 years. Compare the results with the three state multi-state model.

6. Covariates

Time to get back to the study question, which is about the impact of a relapse on survival, and how that has changed over the course of (calendar) time.

We will first make some frequency tables. First of the variable coding the period in which the relapse was detected.

```
> table(ebmt1$yrel,exclude=NULL)

1993-1996 1997-1999     2000-      <NA> 
      103       208       145      1521 
```

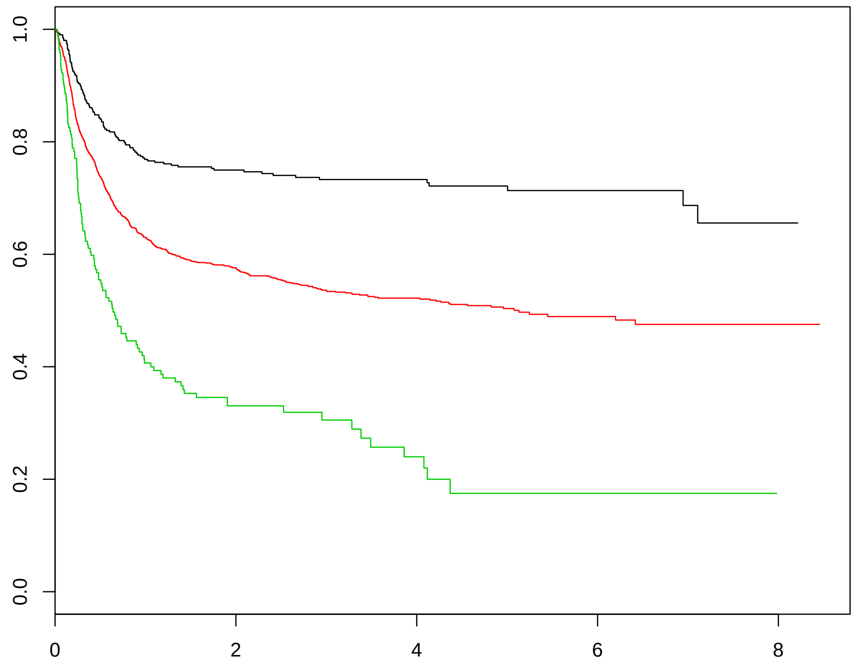

Figure 8: Overall survival Kaplan-Meier curves for the three risk groups

Note that there are 1521 NA's (why?) and a total of 456 patients for which year of relapse is known. Another important variable is the EBMT score, given by `score`.

```
> table(ebmt1$score)

   Low risk Medium risk   High risk
        406        1404         167
```

6.1. Standard analysis

Let us first look at the effect of survival of `score`, the EBMT risk score. First we will make separate Kaplan-Meier survival curves for the three risk groups. This can be done using *survfit*, as in the case without the score groups. The result is shown in Figure 8.

```
> sf3 <- survfit(Surv(srv, srvstat) ~ score, data=ebmt1)
> plot(sf3, xscale=year, col=1:3)
```

It seems clear that the EBMT score has a profound influence on survival. Moreover, the proportional hazards assumption on overall survival with respect to the EBMT score looks reasonable, judging from Figure 8. The hazard ratios can be estimated with the function *coxph*. They indicate a clear effect of EBMT score on survival.

```
> c2 <- coxph(Surv(srv, srvstat) ~ score, data = ebmt1)
> c2

Call:
coxph(formula = Surv(srv, srvstat) ~ score, data = ebmt1)

                  coef exp(coef) se(coef)    z      p
scoreMedium risk 0.684     1.983    0.104  6.6  4e-11
scoreHigh risk   1.377     3.962    0.134 10.3 <2e-16

Likelihood ratio test=106.3  on 2 df, p=<2e-16
n= 1977, number of events= 872
```

Question 14. Locate the hazard ratios. What do they mean?

Question 15. Perform a Cox regression to estimate the effect of EBMT risk group on time to relapse. Locate the hazard ratios and write them down.

6.2. Multi-state model

We will now turn to estimation of the effects of `score` and of the year of relapse in the context of the multi-state model considered in Section 4. Make sure the definitions of the transition matrix `tmat` and the long format data `msebmt` are (still) in memory.

Transition-specific covariates can be defined in the dataset using the function *expand.covs*. The last argument is a vector containing the names of the covariates in the dataset for which transition-specific covariates are to be created. By default these will be included in the resulting dataset, with suffix .1, .2, ..., .K.

```
> msebmtsav <- msebmt
> msebmt <- expand.covs(msebmt, covs)
```

Let us have a look at the result

```
> head(msebmt)

An object of class 'msdata'

Data:
  patid from to trans Tstart Tstop time status       score      yrel
1     1    1  2     1      0  1610 1610      0 Medium risk      <NA>
2     1    1  3     2      0  1610 1610      0 Medium risk      <NA>
3     2    1  2     1      0   165  165      1 Medium risk 1997-1999
4     2    1  3     2      0   165  165      0 Medium risk 1997-1999
5     2    2  3     3    165   961  796      1 Medium risk 1997-1999
6     3    1  2     1      0  1508 1508      0 Medium risk      <NA>
  rel.srv scoreMedium.risk.1 scoreMedium.risk.2 scoreMedium.risk.3
1       0                  1                  0                  0
```

```
2       0                 0                 1                 0
3       0                 1                 0                 0
4       0                 0                 1                 0
5       1                 0                 0                 1
6       0                 1                 0                 0
  scoreHigh.risk.1 scoreHigh.risk.2 scoreHigh.risk.3 yrel1997.1999.1
1                0                0                0              NA
2                0                0                0              NA
3                0                0                0               1
4                0                0                0               0
5                0                0                0               0
6                0                0                0              NA
  yrel1997.1999.2 yrel1997.1999.3 yrel2000..1 yrel2000..2 yrel2000..3
1              NA              NA          NA          NA          NA
2              NA              NA          NA          NA          NA
3               0               0           0           0           0
4               1               0           0           0           0
5               0               1           0           0           0
6              NA              NA          NA          NA          NA
```

Question 16. Study the values of the transition-specific covariates, in relation to the basic covariates and the transitions.

The names in the result are overly long. Often it is preferable to replace the factor levels in the column names by numerical values. The disadvantage is that the user needs to remember what these numbers represent. For example:

```
> msebmt <- msebmtsav
> msebmt <- expand.covs(msebmt, covs, longnames=FALSE)
> head(msebmt)
```

```
An object of class 'msdata'

Data:
  patid from to trans Tstart Tstop time status       score      yrel
1     1    1  2     1      0  1610 1610      0 Medium risk      <NA>
2     1    1  3     2      0  1610 1610      0 Medium risk      <NA>
3     2    1  2     1      0   165  165      1 Medium risk 1997-1999
4     2    1  3     2      0   165  165      0 Medium risk 1997-1999
5     2    2  3     3    165   961  796      1 Medium risk 1997-1999
6     3    1  2     1      0  1508 1508      0 Medium risk      <NA>
  rel.srv score1.1 score1.2 score1.3 score2.1 score2.2 score2.3
1       0        1        0        0        0        0        0
2       0        0        1        0        0        0        0
3       0        1        0        0        0        0        0
4       0        0        1        0        0        0        0
5       1        0        0        1        0        0        0
```

```
6        0        1        0        0        0        0        0
  yrel1.1 yrel1.2 yrel1.3 yrel2.1 yrel2.2 yrel2.3
1      NA      NA      NA      NA      NA      NA
2      NA      NA      NA      NA      NA      NA
3       1       0       0       0       0       0
4       0       1       0       0       0       0
5       0       0       1       0       0       0
6      NA      NA      NA      NA      NA      NA
```

Question 17. What does for instance `score2.3` represent?

For transition 3 (and only for this transition), `yrel` is of importance, but there are `NA` values. We will replace these by 0, since otherwise these rows (transitions 1 and 2) are removed in fitting a Cox model, which is not what we want.

```
> msebmt$yrel1.3[is.na(msebmt$yrel1.3)] <- 0
> msebmt$yrel2.3[is.na(msebmt$yrel2.3)] <- 0
> head(msebmt)

An object of class 'msdata'

Data:
  patid from to trans Tstart Tstop time status       score      yrel
1     1    1  2     1      0  1610 1610      0 Medium risk      <NA>
2     1    1  3     2      0  1610 1610      0 Medium risk      <NA>
3     2    1  2     1      0   165  165      1 Medium risk 1997-1999
4     2    1  3     2      0   165  165      0 Medium risk 1997-1999
5     2    2  3     3    165   961  796      1 Medium risk 1997-1999
6     3    1  2     1      0  1508 1508      0 Medium risk      <NA>
  rel.srv score1.1 score1.2 score1.3 score2.1 score2.2 score2.3
1       0        1        0        0        0        0        0
2       0        0        1        0        0        0        0
3       0        1        0        0        0        0        0
4       0        0        1        0        0        0        0
5       1        0        0        1        0        0        0
6       0        1        0        0        0        0        0
  yrel1.1 yrel1.2 yrel1.3 yrel2.1 yrel2.2 yrel2.3
1      NA      NA       0      NA      NA       0
2      NA      NA       0      NA      NA       0
3       1       0       0       0       0       0
4       0       1       0       0       0       0
5       0       0       1       0       0       0
6      NA      NA       0      NA      NA       0
```

The first model that we will fit is a Cox model, stratified according to `trans`, where the effect of the EBMT score is equal across transitions.

```
> c2 <- coxph(Surv(Tstart,Tstop,status) ~ score + strata(trans),
+     data=msebmt)
> c2

Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ score + strata(trans),
    data = msebmt)

                   coef exp(coef) se(coef)     z       p
scoreMedium risk 0.5459    1.7262   0.0799  6.83 8.4e-12
scoreHigh risk   1.1924    3.2949   0.1076 11.08 < 2e-16

Likelihood ratio test=119.6  on 2 df, p=<2e-16
n= 4410, number of events= 1328
```

Question 18. Locate and interpret the hazard ratios.

We can use the transition-specific covariates to estimate different EBMT score effects across transitions.

```
> c3 <- coxph(Surv(Tstart,Tstop,status) ~ score1.1 + score2.1 +
+     score1.2 + score2.2 + score1.3 + score2.3 + strata(trans),
+     data=msebmt)
> c3

Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ score1.1 + score2.1 +
    score1.2 + score2.2 + score1.3 + score2.3 + strata(trans),
    data = msebmt)

          coef exp(coef) se(coef)    z       p
score1.1 0.432     1.540    0.126 3.44 0.00058
score2.1 1.085     2.961    0.183 5.93 2.9e-09
score1.2 0.639     1.895    0.113 5.64 1.7e-08
score2.2 1.178     3.249    0.153 7.68 1.6e-14
score1.3 0.542     1.720    0.260 2.08 0.03707
score2.3 1.473     4.362    0.301 4.90 9.6e-07

Likelihood ratio test=123.9  on 6 df, p=<2e-16
n= 4410, number of events= 1328
```

This model could in fact be obtained as well by fitting separate Cox regression models for each transition, for instance by

```
> coxph(Surv(Tstart,Tstop,status) ~ score, data=msebmt,
+     subset=(trans==1))
```

```
Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ score, data = msebmt,
    subset = (trans == 1))

                  coef exp(coef) se(coef)    z       p
scoreMedium risk 0.432     1.540    0.126 3.44 0.00058
scoreHigh risk   1.085     2.961    0.183 5.93 2.9e-09

Likelihood ratio test=33.03  on 2 df, p=7e-08
n= 1977, number of events= 456
```

Question 19. Fit the Cox models for the other two transitions and see whether they correspond to the results in `c3`.

Question 20. Looking at the hazard ratios of medium and high risk EBMT scores across the different transitions, would you say they are similar or not?

The hypothesis that these hazard ratios are equal across transitions can be tested with a likelihood ratio test. This is done using the function *anova*.

```
> anova(c2,c3)
```

```
Analysis of Deviance Table
 Cox model: response is  Surv(Tstart, Tstop, status)
 Model 1: ~ score + strata(trans)
 Model 2: ~ score1.1 + score2.1 + score1.2 + score2.2 + score1.3 + score2.3 + strata(trans
   loglik  Chisq Df P(>|Chi|)
1 -8976.2
2 -8974.1 4.2774  4    0.3698
```

Question 21. What is your conclusion with regard to the hypothesis above?

We will stick with a model with the same effect of EBMT score across transitions.

We will look now at a model where the baseline transition intensities into death are assumed to be proportional, as was done in Section 5. Again, we do this by defining a time-dependent covariate `rel.surv` which is 0 for transition 2 and 1 for transition 3, so it is 0 if no relapse has occurred and 1 if a relapse has occurred. It measures the effect of relapse on survival. The whole model can be fitted (adjusted for EBMT score) using a stratified Cox regression model with strata defined by the receiving state (`to` in the data), and with `rel.srv` as additional covariate.

```
> msebmt$rel.srv <- 0
> msebmt$rel.srv[msebmt$trans==3] <- 1
> c4 <- coxph(Surv(Tstart,Tstop,status) ~ score + rel.srv +
+     strata(to), data=msebmt)
> c4
```

```
Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ score + rel.srv +
    strata(to), data = msebmt)

                   coef exp(coef) se(coef)     z      p
scoreMedium risk 0.5451    1.7248   0.0799  6.82  9e-12
scoreHigh risk   1.1920    3.2937   0.1076 11.08 <2e-16
rel.srv          1.0549    2.8718   0.0893 11.82 <2e-16

Likelihood ratio test=255.9  on 3 df, p=<2e-16
n= 4410, number of events= 1328
```

Question 22. What does the hazard ratio of `rel.srv` imply?

Question 23. Have the hazard ratios of `score` (or their standard errors) changed compared to c2?

Now we are ready to turn to the actual study question, whether the effect of a relapse has changed over time. We will start from the last model with proportional baseline hazards for the two transitions into death. We now add `yrel` to the model.

```
> c5 <- coxph(Surv(Tstart,Tstop,status) ~ score + yrel1.3 + yrel2.3 +
+     rel.srv + strata(to), data=msebmt)
> c5

Call:
coxph(formula = Surv(Tstart, Tstop, status) ~ score + yrel1.3 +
    yrel2.3 + rel.srv + strata(to), data = msebmt)

                   coef exp(coef) se(coef)     z       p
scoreMedium risk  0.542     1.720    0.080  6.78 1.2e-11
scoreHigh risk    1.202     3.326    0.108 11.17 < 2e-16
yrel1.3          -0.163     0.850    0.164 -0.99 0.32022
yrel2.3          -0.844     0.430    0.240 -3.52 0.00043
rel.srv           1.300     3.670    0.140  9.28 < 2e-16

Likelihood ratio test=270.7  on 5 df, p=<2e-16
n= 4410, number of events= 1328
```

Question 24. What is your conclusion with regard to the effect of period of relapse?

Question 25. Has the hazard ratio of `rel.srv` changed? Has the *meaning* of this hazard ratio changed?

6.3. Estimation of hazards

The **mstate** software contains a function *`msfit`* that can be used to estimate the (patient-specific) cumulative hazards for all transitions in the multi-state model corresponding to a patients with a given set of covariate values. It is very similar to *survfit*, which can do the same in principle, but tuned to its use in multi-state models. To use it, you need to specify a `newdata` data frame, as for *survfit*, containing at least the covariate values of the patient for all covariates used in the Cox model on which it is based, and with the same factor levels and labels, if used (!). For use in *msfit*, the `newdata` data frame has to have K rows (K being the number of transitions), a column `trans` numbered 1 through K, and a `strata` column, specifying for each transition to which stratum in the Cox model it corresponds; the numbering of strata is as in the *coxph* object. Here with *strata=c(1,2,2)* we have specified that in the multi-state model the first transition corresponds to the first stratum (`to`=2) in the *coxph* object `c5`, and that transitions 2 and 3 both correspond to the second stratum (`to`=3) in `c5`. Often it is convenient to start with the overall covariates and to mimic the relevant steps of the data building process in this new dataset (in particular *expand.covs* can be used). Since we don't have many covariates, it is quicker to define `ndata` directly. We do this now for a patient whose covariates are the reference values; this way we obtain the estimated baseline transition hazards.

```
> ndata <- data.frame(trans=1:3, from=c(1,1,2),
+     to=c(2,3,3), score=1)
> ndata$score <- factor(ndata$score, levels=1:3,
+     labels=levels(msebmt$score))
> ndata$strata <- c(1,2,2)
> ndata$yrel2.3 <- ndata$yrel1.3 <- 0
> ndata$rel.srv <- 0
> ndata$rel.srv[ndata$trans==3] <- 1
> ndata

  trans from to    score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 Low risk      1       0       0       0
2     2    1  3 Low risk      2       0       0       0
3     3    2  3 Low risk      2       0       0       1
```

Now we call *msfit*.

```
> HvH <- msfit(c5,newdata=ndata,trans=tmat)
```

The resulting object is an object of class `"msfit"`, which is basically a list with two components, the first, `Haz`, containing the estimated patient-specific cumulative hazard for each of the transitions in the multi-state model; the second, `varHaz`, containing the covariances of the estimated patient-specific cumulative hazards for each pair of transitions. Those cumulative hazards and their covariances for each (pair of) transitions are evaluated at each unique event time point (for any event). As such, it contains too much information, but it is needed for subsequent use in the Aalen-Johansen estimator. There is a *summary* method for `"msfit"` objects.

```
> summary(HvH)

An object of class 'msfit'
```

```
Transition 1 (head and tail):

  time          Haz trans
1  0.5 0.0005924340     1
2  1.0 0.0008891803     1
3  2.0 0.0008891803     1
4  5.0 0.0008891803     1
5  7.0 0.0008891803     1
6  8.0 0.0008891803     1

...

    time       Haz trans
597 2322 0.3195808     1
598 2344 0.3195808     1
599 2537 0.3195808     1
600 2596 0.3195808     1
601 2597 0.3391839     1
602 3088 0.3391839     1

Transition 2 (head and tail):

    time          Haz trans
603  0.5 0.0005924340     2
604  1.0 0.0005924340     2
605  2.0 0.0008878371     2
606  5.0 0.0017747592     2
607  7.0 0.0023667933     2
608  8.0 0.0029593994     2

...

     time       Haz trans
1199 2322 0.3362109     2
1200 2344 0.3397340     2
1201 2537 0.3457876     2
1202 2596 0.3536825     2
1203 2597 0.3536825     2
1204 3088 0.3536825     2

Transition 3 (head and tail):

     time         Haz trans
1205  0.5 0.002174353     3
1206  1.0 0.002174353     3
1207  2.0 0.003258542     3
1208  5.0 0.006513725     3
```

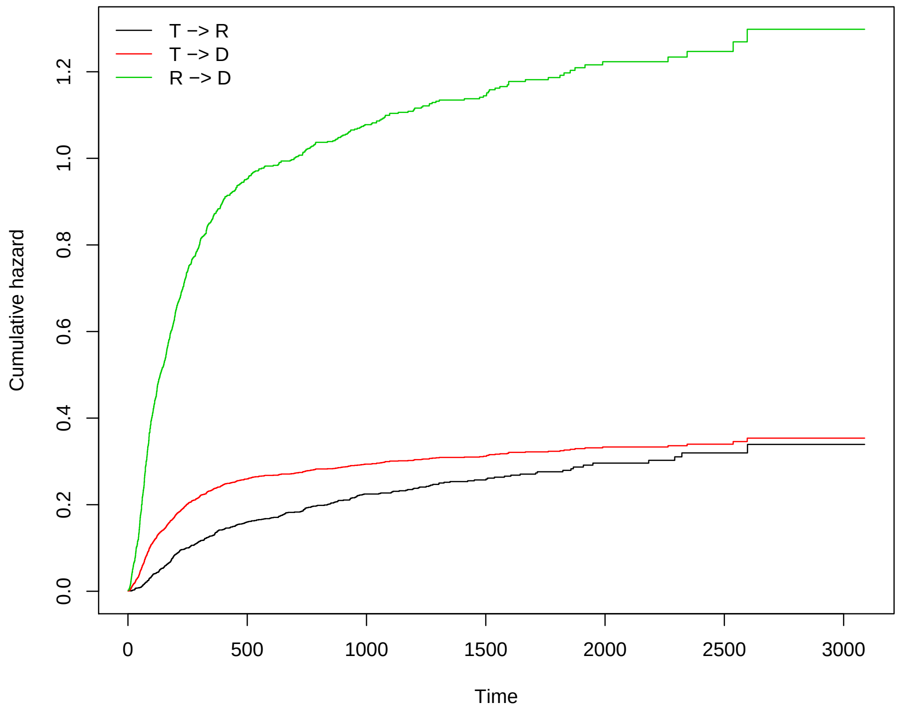

Figure 9: Cumulative baseline hazards for the proportional baselines model

```
1209   7.0 0.008686610     3
1210   8.0 0.010861594     3

...

     time      Haz trans
1801 2322 1.233962     3
1802 2344 1.246893     3
1803 2537 1.269110     3
1804 2596 1.298086     3
1805 2597 1.298086     3
1806 3088 1.298086     3
```

By default it gives the head and tail of the hazard estimates of each of the transitions. There is also a ***plot*** method for `"msfit"` objects. Figure 9 shows a plot of the baseline cumulative transition hazards.

```
> plot(HvH)
```

Question 26. The hazards of transitions 2 and 3 (from transplant to death and from relapse to death, respectively) are given by

```
> H0 <- HvH$Haz[HvH$Haz$trans==2,]
> H1 <- HvH$Haz[HvH$Haz$trans==3,]
```

What happens if you look at the ratio between these two hazards?

```
> head(H1$Haz/H0$Haz)

[1] 3.670202 3.670202 3.670202 3.670202 3.670202 3.670202
```

We now try to get a visual impression of the effect of period of relapse on the hazard of death after relapse. For the EBMT score we retain the reference category (low risk). We have already defined `H0` and `H1` as the cumulative hazards for transplant to death and relapse to death for the reference relapse period. To obtain the other two hazards corresponding to period of relapse 1997-1999 and 2000+, we multiply the hazard in `H1` with the hazard ratios of the two other periods with respect to the first. These hazard ratios can be extracted from the `coef` item of the *coxph* object. Alternatively, we could rebuild `newdata` datasets and use *msfit* again. First the code to define the hazards;

```
> H2 <- H3 <- H1
> H2$Haz <- H2$Haz*exp(c5$coef[3])
> H3$Haz <- H3$Haz*exp(c5$coef[4])
```

The result is shown in Figure 10.

```
> plot(H1$time/365.25,H1$Haz,type="s",ylim=c(0,max(H1$Haz)),
+     xlab="Years since transplant",ylab="Cumulative hazard",
+     lwd=2,col="red2")
> lines(H2$time/365.25,H2$Haz,type="s",lwd=2,col="orangered")
> lines(H3$time/365.25,H3$Haz,type="s",lwd=2,col="orange")
> lines(H0$time/365.25,H0$Haz,type="s",lwd=2,col=3)
> legend("topleft",
+     c("Relapse in 1993-1996","Relapse in 1997-1999",
+         "Relapse in 2000 or later","No relapse"),
+     lwd=2,col=c("red2","orangered","orange",3),bty="n")
```

6.4. Estimation of probabilities

The impact of period of relapse on survival looks quite impressive, but the question remains how large the impact is for a population followed from transpantation; after all, if hardly anyone gets a relapse, large difference in mortality after relapse will not have a big impact in a population after transplantation. In order to study this we need to be able to compute the probability of dying after transplantation, with or without relapse, for the three different relapse periods. We will start with the first period relapse, since we have already done some preliminary work for this.

Question 27. What does a prediction probability of death starting from transplantation for a patient with a relapse between 1993 and 1996 actually mean?

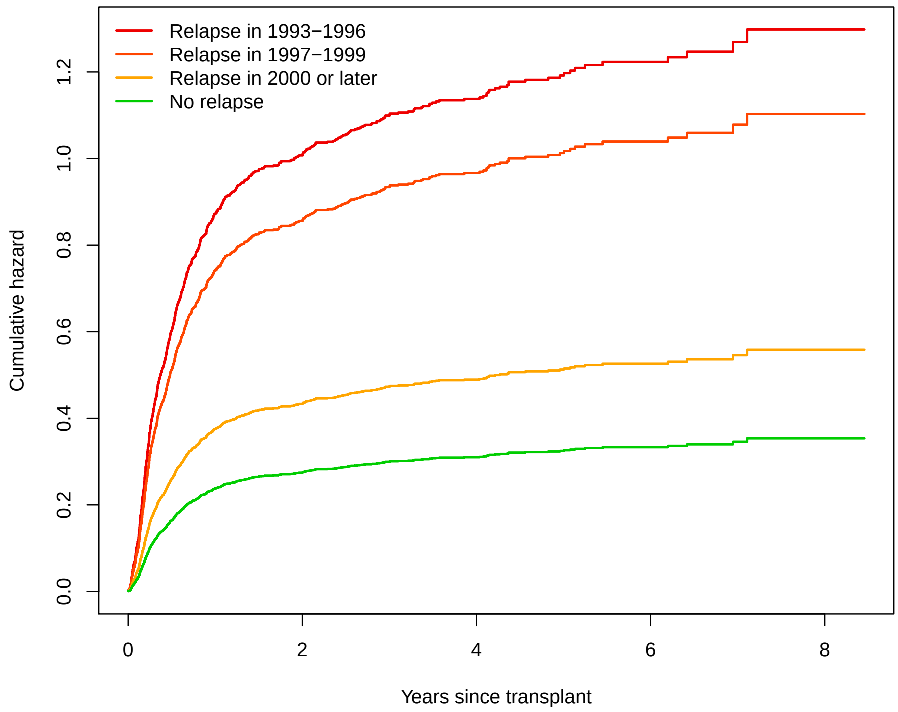

Figure 10: Cumulative hazards for death without relapse and with relapse in three periods

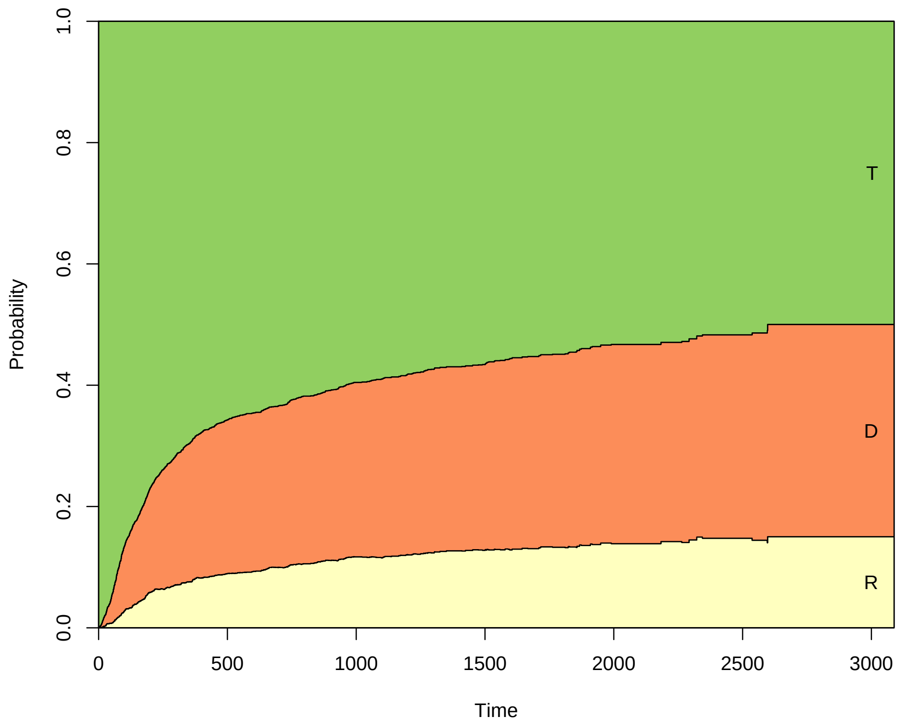

Figure 11: Stacked (and filled) prediction probabilities from state 1 at time=0, for a low risk patient transplanted in 1993-1996

Question 28. How can we justify this?

The result of *msfit* can be used as input for *probtrans*, as illustrated in Section 5. We will call *probtrans* with HvH1 as input.

```
> HvH1 <- HvH
> pt1 <- probtrans(HvH1,predt=0,direction="forward")
```

We will show the model-based transition probabilities in Figure 11, using a "filled" plot.

```
> plot(pt1, ord=c(2,3,1), type="filled")
```

Now we will concentrate on survival probabilities and compare the predicted probabilities we just obtained (patients having a relapse in 1993-1996) with those obtained from patients having a relapse in 1997-1999, or 2000+ (and with the same EBMT risk score, low risk). In order to obtain these prediction probabilities we repeat what we have just done, changing only the value of yrel where necessary. The code is tedious, but straightforward.

```
> ndata.copy <- ndata # make a copy for later
> # first dummy variable = 1 for transition 3
> ndata$yrel1.3[ndata$trans==3] <- 1
> ndata
```

```
  trans from to    score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 Low risk      1       0       0       0
2     2    1  3 Low risk      2       0       0       0
3     3    2  3 Low risk      2       1       0       1
```

```
> HvH2 <- msfit(c5,newdata=ndata,trans=tmat)
> ndata <- ndata.copy # use the copy
> # second dummy variable = 1 for transition 3
> ndata$yrel2.3[ndata$trans==3] <- 1
> ndata
```

```
  trans from to    score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 Low risk      1       0       0       0
2     2    1  3 Low risk      2       0       0       0
3     3    2  3 Low risk      2       0       1       1
```

```
> HvH3 <- msfit(c5,newdata=ndata,trans=tmat)
> pt11 <- pt1[[1]]
> pt2 <- probtrans(HvH2,predt=0,direction="forward")
> pt21 <- pt2[[1]]
> pt3 <- probtrans(HvH3,predt=0,direction="forward")
> pt31 <- pt3[[1]]
```

The result is shown in Figure 12.

```
> plot(pt11$time/365.25,pt11$pstate3,type="s",ylim=c(0,1),
+     xlab="Years since transplant",ylab="Probability of death",
+     lwd=2,col="red2")
> lines(pt21$time/365.25,pt21$pstate3,type="s",lwd=2,col="orangered")
> lines(pt31$time/365.25,pt31$pstate3,type="s",lwd=2,col="orange")
> legend("topleft",
+     c("Relapse in 1993-1996","Relapse in 1997-1999",
+         "Relapse in 2000 or later"),
+     lwd=2,col=c("red2","orangered","orange"),bty="n")
```

Question 29. What is your conclusion from Figure 12?

Question 30. Would you expect the impact of period of relapse on survival after transplant to be stronger for a high risk patient?

Question 31. Make a plot similar to Figure 12 for a high risk patient.

Here is the code:

```
> ndata <- ndata.copy # use copy from last time
> ndata$score <- "High risk"
> ndata
```

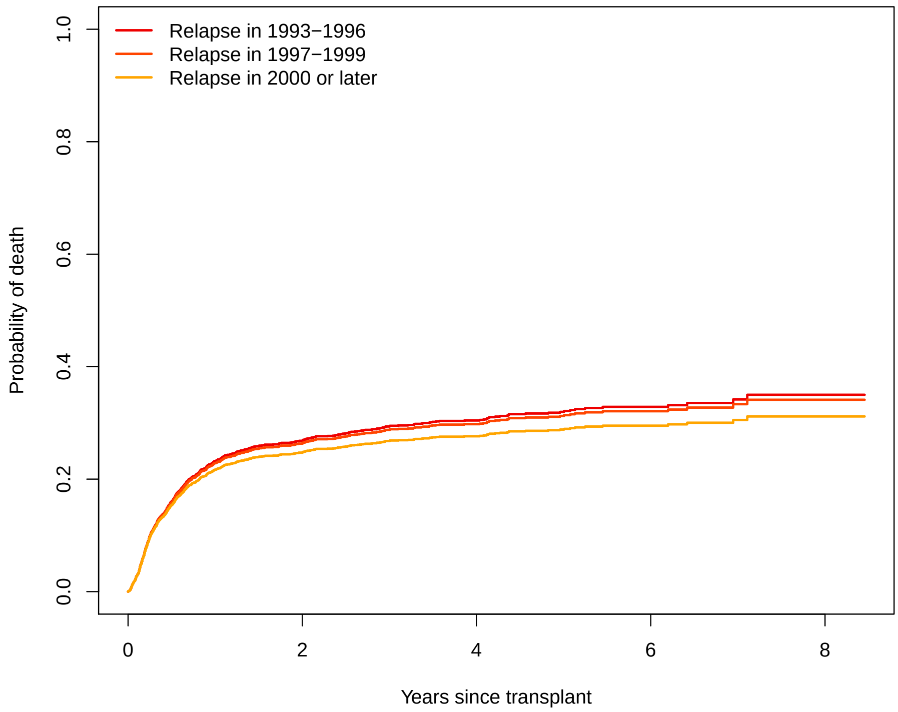

Figure 12: Predictions from state 1 at time=0 for a low risk patient

```
  trans from to     score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 High risk      1       0       0       0
2     2    1  3 High risk      2       0       0       0
3     3    2  3 High risk      2       0       0       1

> ndata.copy <- ndata # make a copy for later
> HvH1 <- msfit(c5,newdata=ndata,trans=tmat)
> ndata <- ndata.copy # used the copy
> # first dummy variable = 1 for transition 3
> ndata$yrel1.3[ndata$trans==3] <- 1
> ndata

  trans from to     score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 High risk      1       0       0       0
2     2    1  3 High risk      2       0       0       0
3     3    2  3 High risk      2       1       0       1

> HvH2 <- msfit(c5,newdata=ndata,trans=tmat)
> ndata <- ndata.copy
> # second dummy variable = 1 for transition 3
> ndata$yrel2.3[ndata$trans==3] <- 1
> ndata

  trans from to     score strata yrel1.3 yrel2.3 rel.srv
1     1    1  2 High risk      1       0       0       0
2     2    1  3 High risk      2       0       0       0
3     3    2  3 High risk      2       0       1       1

> HvH3 <- msfit(c5,newdata=ndata,trans=tmat)
> pt1 <- probtrans(HvH1,predt=0,direction="forward")
> pt11 <- pt1[[1]]
> pt2 <- probtrans(HvH2,predt=0,direction="forward")
> pt21 <- pt2[[1]]
> pt3 <- probtrans(HvH3,predt=0,direction="forward")
> pt31 <- pt3[[1]]
```

And the plot is shown in Figure 13.

```
> plot(pt11$time/365.25,pt11$pstate3,type="s",ylim=c(0,1),
+     xlab="Years since transplant",ylab="Probability of death",
+     lwd=2,col="red2")
> lines(pt21$time/365.25,pt21$pstate3,type="s",lwd=2,col="orangered")
> lines(pt31$time/365.25,pt31$pstate3,type="s",lwd=2,col="orange")
> legend("topleft",
+     c("Relapse in 1993-1996","Relapse in 1997-1999",
+         "Relapse in 2000 or later"),
+     lwd=2,col=c("red2","orangered","orange"),bty="n")
```

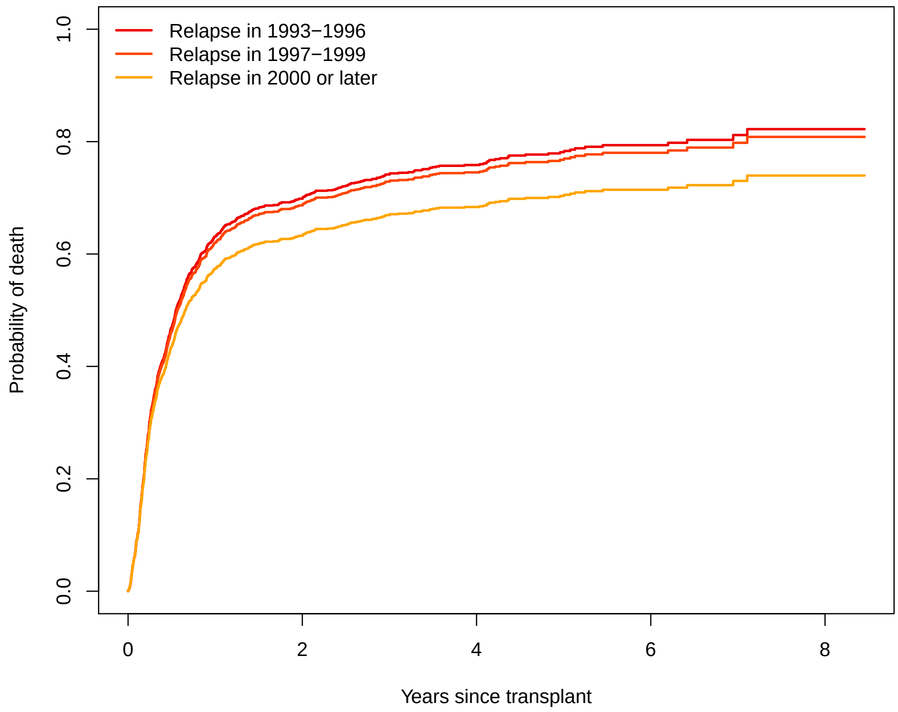

Figure 13: Predictions from state 1 at time=0 for a high risk patient

Bonus Question 1. Go back to the Cox model `c5`. What is the hazard ratio of the transition into death after relapse in 2000+ with respect to the transition into death without relapse? Does a relapse nowadays (after 2000) have a (statistically) significant impact on survival?

Bonus Question 2. Based on this model, how would you calculate the probability of having died after relapse for, say, a low risk subject?

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